Alex Schutz

alex-schutz.github.io

EDUCATION

University of Oxford — *DPhil Engineering Science (Robotics)*

OCTOBER 2023 - PRESENT

PhD programme with the Oxford Robotics Institute focussing on multi-agent planning in uncertain environments. Supervisors: Nick Hawes and Bruno Lacerda.

- Designed novel algorithm for solving goal-driven deterministic POMDPs (IJCAI 2025)
- Currently investigating Neural Algorithmic Reasoning in NP-hard problem contexts
- Working on Graph Reinforcement Learning for multi-agent topological planning problems

Key areas: Reinforcement Learning, Graph Neural Networks, POMDPs, Multi-Agent Planning

University of Adelaide — *B.Eng (Hons)(Mechatronics) w B.MaCSci (Pure Maths)*

JANUARY 2016 - DECEMBER 2020

Awarded First Class Honours and achieved a GPA of 6.919/7.
Key subjects: Robotics, Control Systems, Microcontroller Programming, Groups & Rings, Real Analysis

University of Adelaide — *Honours Project*

MARCH 2020 - NOVEMBER 2020

Completed research project titled Modelling Mining Trucks for Hauling Predictions. Trained and evaluated different machine learning models using scikit-learn. Simulated model performance in non-deterministic mine site model. Served as technical lead in a team of four.
Key areas: Machine Learning, Analysis, Project Management

EXPERIENCE

Myriota — *Rapid Prototyping Engineer*

JANUARY 2021- SEPTEMBER 2023

Development of algorithms and software for the operation of satellite hardware in IoT applications.

- Designed software tools using C/C++ and Python for large scale cloud-based simulations
- Analysed data from simulation to determine optimal satellite network configurations
- Architected software solutions to coordinate radio transmissions and receptions on a satellite
- Designed PCB layouts and developed FPGA applications for real-time decoding

PUBLICATIONS

International Joint Conference on Artificial Intelligence (IJCAI) 2025 | arXiv:2505.00596
A Finite-State Controller Based Offline Solver for Deterministic POMDPs
A Schutz, Y You, M Mattamala, I Caliskanelli, B Lacerda, N Hawes

Preprint (Submitted to ICLR 2026) | arXiv:2509.18930
Tackling GNARLy Problems: Graph Neural Algorithmic Reasoning Reimagined through Reinforcement Learning
A Schutz, VA Darvariu, E Panagiotaki, B Lacerda, N Hawes

AWARDS

General Sir John Monash Scholar for the year 2023
The General Sir John Monash Foundation awards postgraduate scholarships to Australians who demonstrate outstanding leadership potential and who wish to study overseas.

University of Adelaide

- Valedictorian for Mechanical Engineering *2021*
- Frederic Barnes Waldron Best Student Award *2021*
- Mechatronic Engineering Student Prize *2021*
- Most Outstanding Engineering Student Prize *2021*
- R.J. Jennings Memorial Prize for Mechanical Engineering Design Prize *2021*
- Nancy Webb Prize in Mathematics *2019*
- Andy Thomas Scholarship *2016*

LEADERSHIP EXPERIENCE

Team Lead for Robotics Inclusive *2024 - Present*

- Partnering with organisations to promote inclusive practices in the robotics industry
- Production of a new website and cohesive brand image

Treasurer/Secretary for the Oxford Robotics Graduate Society *2024 - Present*

- Founding treasurer of the organisation
- Planning regular events for 50+ students within the Oxford Robotics Institute

SKILLS

- Programming: C/C++, Python, MATLAB
- Machine learning: Deep learning, Reinforcement learning, Supervised learning, Graph Neural Networks
- Frameworks: PyTorch, JAX, SB3, WandB
- Organisation: Operationalisation, Event planning, Teaching